

Amy Kuang

Riverside Area, CA | (951) 546-4767 | amy6kuang@gmail.com | www.linkedin.com/in/kuanga

Education

University of California Irvine | Master of Engineering, Mechanical & Aerospace Engineering

M.Eng. May 2027

Major Area: Autonomous and Intelligent Machine & Systems

Colorado School of Mines | Bachelor of Science, Mechanical Engineering

B.S. May 2026

Focus: Manufacturing | GPA: 3.0

Technical Skills

Platforms: SolidWorks | LabVIEW | Arduino IDE | MATLAB | Microsoft Office | PrusaSlicer | Ideamaker | Python | CorelDraw

Technical: Lean Six-Sigma | Additive Manufacturing | Machine Shop Proficiency | Finite Element Analysis | Model-Based Systems

Personal: Team-Based Leadership | Adaptability | Attention to Detail | Cross-Functional Communication | Positive Influence | Desire to Learn | Conflict Resolution

Professional Experience

CAM Supply Inc. → Design Engineering Intern

Lake Elsinore, CA

(May 2026 – Present)

- Designed and fabricated supporting parts for tattoo pens and acupuncture products via 3D printing (SolidWorks/IdeaMaker)
- Generated engineering drawings with dimensions, tolerances, and manufacturing specifications to support production and quality control
- Reviewed and updated product documentation to maintain compliance with regulatory and quality requirements, including FDA product records
- Conducted cost-benefit analyses of U.S. and Chinese manufacturing partners, evaluating labor, material, logistics, and production costs to support sourcing decisions

Niagara Bottling LLC. → Manufacturing Engineer Intern

Denver, CO

Niagara Bottling

Denver, Colorado

- Improved production capacity by 420,800 bottles per peak production period by analyzing pack changeover operations and improving process efficiency
- Applied SMED principles to reduce average pack changeover time by 12% (17 → 15 min), directly improving OEE and resulting in a 0.139% throughput efficiency gain
- Standardized MES fault codes and implemented automated aggregate unit tracking to support Root Cause Corrective Action (RCCA) investigations
- Collaborated with maintenance, production, and quality teams to troubleshoot line inefficiencies and improve overall operations; developed SOPs for line improvement and factory equipment to improve repeatability

Field Related Projects

Finite Element Analysis of a Double Wishbone Suspension System

December 2024

- Conducted comprehensive Finite Element Analysis (FEA) to perform a static optimization study minimizing component mass while adhering to safety factor standards
- Evaluated stress distribution and placement under various loading conditions to ensure reliability in high-performance environments; generated technical documentation summarizing trade-offs and validation results

PID-Controlled Easy Bake Oven

November 2024 – December 2024

- Designed a closed-loop electro-mechanical control system using thermocouple feedback and LabVIEW; Engineered a MOSFET-based control circuit bridging digital software logic and analog heating elements
- Performed system integration, hardware-software validation, and functional testing; developed a custom UI to adjust PID gains and monitor system performance via live waveform charts

Embedded Electronics in Additively Manufactured Components

August 2025 – May 2026

Senior Capstone Research Project – Sponsor: Honeywell Internation, Inc. | Colorado School of Mines

Golden, Colorado

- Researched feasibility of embedding functional thermocouples and strain gauges within structural components using a novel multi-material Directed Energy Deposition (DED) process, serving as proof-concept for embedded sensing in aerospace structures
- Developed a hybrid powder deposition technique to safely fuse LPBF-grade Cu and CuNi30 powders using DED laser remelting within stainless-steel channels, resolving an airborne particulate explosion risk identified mid-project
- Translated Honeywell customer requirements into ASTM-compliant coupon designs, drafted test plans, qualification procedures, and a final design report with recommended next steps for future DED system acquisitions

- Investigated powder flow stability in DED systems using combined CFD modeling and transparent-geometry experimental imaging to characterize carrier gas behavior and particle transport in a novel electrodynamic feeder
- Modeled argon carrier gas flow through a multi-outlet powder diffuser in Ansys Fluent and identified low-velocity stagnation zones near outlet regions as probable sites for particle settling and feed instability
- Identified a critical gap in multiphase CFD workflows where DPM particle tracking became numerically unstable in multi-outlet geometries due to absent boundary condition guidelines; proposed CFD-DEM coupling as a path forward